

Setting Up a New Puzzle

To create a new, scrambled puzzle, we cannot simply put the tiles in random places, because some configurations of tiles produce invalid, unsolvable puzzles. Instead, we need to start from a solved puzzle and make many random moves. Solving the puzzle becomes a matter of figuring out which slides will undo these random slides to get back to the original, ordered configuration.

But it's not always possible to make moves in each of the four directions. For example, if the blank space is in the bottom-right corner, as in [Figure 12-6](#), tiles can slide only down or right because no tiles can slide left or up. Furthermore, if sliding the 7 tile in [Figure 12-6](#) up was the previous move, then sliding it down is removed as a valid move because it would undo the previous move.

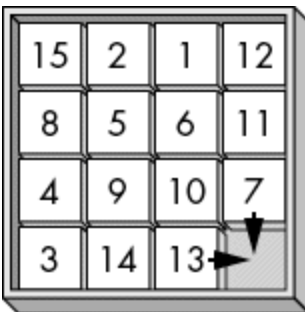


Figure 12-6: If the blank space is in the bottom-right corner, down and right are the only valid slide directions.

To help us, we need a `getValidMoves()` function that can tell us which slide directions are possible on a given board data structure:

Python

```
def getValidMoves(board, prevMove=None):
    """Returns a list of the valid moves to make on this board. If
    prevMove is provided, do not include the move that would undo it."""

    blankx, blanky = findBlankSpace(board)

    validMoves = []
```